Computer Networks

Module 1

Overview of the Internet

Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
  - World Wide Web and HTTP
  - FTP
  - Electronic Mail
  - DNS
- Peer-to-peer paradigm
  - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

TCP/IP Protocol Suite

- TCP/IP stands for:
  - Transmission Control Protocol
  - Internet Protocol
- TCP/IP is:
  - A protocol suite
  - A set of protocols organized in layers
- Used in:
  - The Internet
- TCP/IP is a:
  - Hierarchical protocol
- Made up of:
  - Interactive modules
- Each module provides:
  - Specific functionality

Hierarchical Nature of TCP/IP

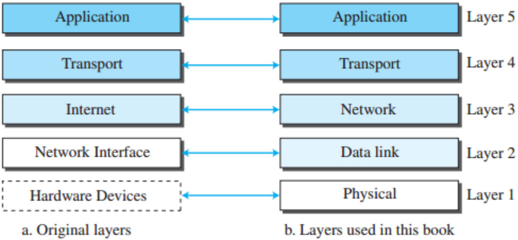
- Hierarchical protocol means:
  - Upper layers depend on lower layers
- Each upper-level protocol:
  - Uses services of lower-level protocols
- Layered design supports:
  - Modularity
  - Flexibility
  - Efficient communication
- Interaction occurs:
  - Between adjacent layers

TCP/IP Layer Configurations

- Original TCP/IP model:
  - Four software layers
  - Built upon hardware
- Modern TCP/IP model:
  - Five-layer model
- Five-layer model provides:
  - Better representation of networking functions
- Figure 1.12 shows:
  - Four-layer configuration
  - Five-layer configuration

Layers in the TCP/IP protocol suite

Figure 1.12 *Layers in the TCP/IP protocol suite*



Layered Architecture in TCP/IP

- TCP/IP communication occurs through:
  - Multiple protocol layers
- Example internet consists of:
  - Three LANs (links)
  - Link-layer switches
  - One router
- Communication occurs between:
  - Computer A
  - Computer B
- Total communicating devices:
  - Source host
  - Link-layer switch in Link 1
  - Router
  - Link-layer switch in Link 2
  - Destination host

Communication through an internet

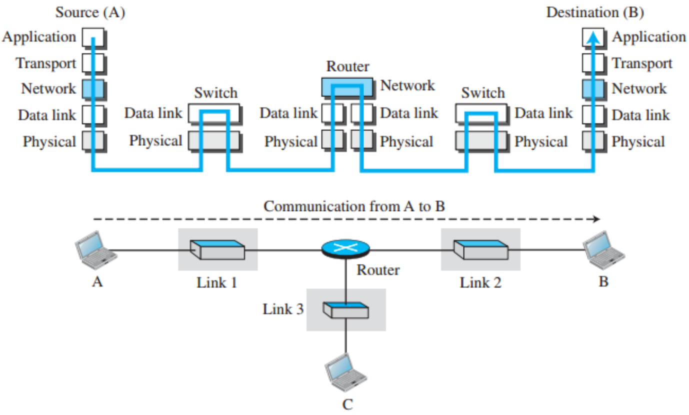


Figure 1.13 *Communication through an internet*

Role of Hosts in Layered Architecture

- Source and destination hosts use:
  - All five TCP/IP layers
- Source host responsibilities:
  - Create message at application layer
  - Pass message through lower layers
  - Physically transmit data
- Destination host responsibilities:
  - Receive data at physical layer
  - Deliver data upward through layers
  - Reach application layer

Router in TCP/IP Architecture

- Router participates in:
  - Network layer
  - Data-link layer
  - Physical layer
- Router does not use:
  - Transport layer
  - Application layer
- Router functionality:
  - Routing packets between networks
- Router connected to:
  - Multiple links
- Each link may use:
  - Different data-link protocols
  - Different physical protocols

Router Handling Different Protocols

- Router receives packet from:
  - One link using one protocol pair
- Router forwards packet to:
  - Another link using another protocol pair
- Example:
  - Packet travels from Link 1 to Link 2
- Router adapts communication between:
  - Different link-layer protocols
  - Different physical-layer protocols

Link-Layer Switch in TCP/IP Architecture

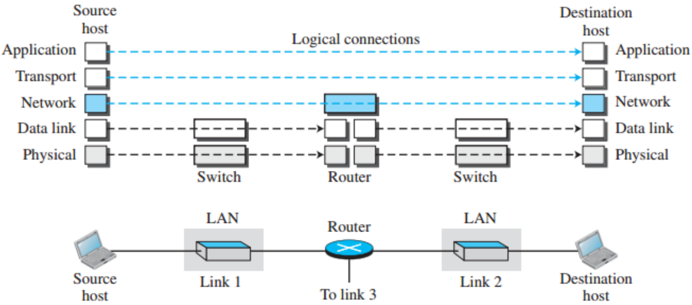
- Link-layer switch operates in:
  - Data-link layer
  - Physical layer
- Switch does not use:
  - Network layer
  - Transport layer
  - Application layer
- Connections of a switch:
  - Belong to same link
- Same link uses:
  - One set of protocols
- Switch handles:
  - One data-link protocol
  - One physical-layer protocol

Layers in the TCP/IP Protocol Suite

- TCP/IP consists of:
  - Multiple protocol layers
- Each layer has:
  - Specific functions
  - Specific duties
- Understanding layers requires:
  - Logical connections between layers
- Logical connections simplify:
  - Understanding of layer responsibilities

Logical connections between layers of the TCP/IP protocol suite

Figure 1.14 *Logical connections between layers of the TCP/IP protocol suite*



End-to-End and Hop-to-Hop Duties

- Application layer duties are:
  - End-to-end
- Transport layer duties are:
  - End-to-end
- Network layer duties are:
  - End-to-end
- Data-link layer duties are:
  - Hop-to-hop
- Physical layer duties are:
  - Hop-to-hop
- A hop refers to:
  - A host
  - A router

Domain of Duties in TCP/IP Layers

- Top three layers:
  - Application layer
  - Transport layer
  - Network layer
- Domain of top three layers:
  - Entire internet
- Bottom two layers:
  - Data-link layer
  - Physical layer
- Domain of bottom two layers:
  - Individual link

Data Units in TCP/IP Layers

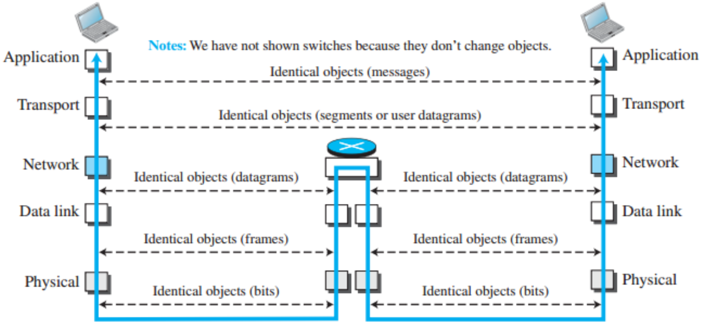
- Each layer creates:
    - A data unit
- In top three layers:
    - Packets remain unchanged
    - Routers do not modify packets
    - Link-layer switches do not modify packets
- In bottom two layers:
    - Packets may change at routers
    - Link-layer switches do not change packets

Identical Objects in Protocol Layering

- Corresponding layers use:
  - Identical objects
- Objects exist below:
  - Each layer
  - Each communicating device
- Logical connections help maintain:
  - Consistency of communication
- Principle ensures:
  - Compatibility between devices

Identical objects in the TCP/IP protocol suite

Figure 1.15 *Identical objects in the TCP/IP protocol suite*



Network Layer and Fragmentation

- Logical connection at network layer exists:
  - Between source host and destination host
- Identical objects exist:
  - Between two hops
- Reason:
  - Routers may fragment packets
- Fragmentation may:
  - Create multiple packets
  - Change number of packets transmitted
- Links between hops:
  - Do not change the object

Description of Each Layer in TCP/IP

- Logical communication helps us understand:
  - Functions of each TCP/IP layer
  - Duties of each layer
- Each layer performs:
  - Specific communication tasks
- Layers work together to provide:
  - End-to-end communication
- TCP/IP layers cooperate through:
  - Logical connections
  - Layered services
- Next sections discuss:
  - Duties of individual layers
  - Functions performed at each layer

Application Layer in TCP/IP

- Application layer communication is:
  - End-to-end
- Two application layers communicate:
  - As though directly connected
- Actual communication occurs through:
  - All TCP/IP layers
- Communication occurs between:
  - Two processes
  - Two programs running at application layer

Duties of the Application Layer

- Application layer provides:
  - Process-to-process communication
- Communication process:
  - One process sends a request
  - Other process sends a response
- Internet application layer contains:
  - Many predefined protocols
- Users can also:
  - Create custom application processes

HTTP and SMTP Protocols

- HTTP:
  - Hypertext Transfer Protocol
  - Used for accessing the World Wide Web (WWW)
- SMTP:
  - Simple Mail Transfer Protocol
  - Main protocol used in e-mail service

FTP, TELNET, and SSH

- FTP:
  - File Transfer Protocol
  - Transfers files between hosts
- TELNET:
  - Terminal Network
  - Used for remote access
- SSH:
  - Secure Shell
  - Secure remote access protocol

SNMP, DNS, and IGMP

- SNMP:
  - Simple Network Management Protocol
  - Used for Internet management
  - Supports local and global administration
- DNS:
  - Domain Name System
  - Finds network-layer address of computers
- IGMP:
  - Internet Group Management Protocol
  - Collects group membership information

Transport Layer in TCP/IP

- Transport layer communication is:
  - End-to-end
- Source transport layer:
  - Receives message from application layer
  - Encapsulates message into transport-layer packet
- Transport-layer packet may be:
  - Segment
  - User datagram
- Packet is logically delivered to:
  - Transport layer at destination host

Duties of the Transport Layer

- Transport layer provides:
  - Services to application layer
- Main responsibility:
  - Deliver application message end-to-end
- Message transfer occurs:
  - Between corresponding application programs
- Transport layer is:
  - Independent of application layer
- Multiple transport protocols exist:
  - Different applications use suitable protocols

Transmission Control Protocol (TCP)

- TCP stands for:
  - Transmission Control Protocol
- TCP is:
  - Connection-oriented protocol
- TCP first establishes:
  - Logical connection between hosts
- TCP creates:
  - Logical pipe for byte stream transfer
- TCP supports:
  - Reliable communication

Services Provided by TCP

- TCP provides:
  - Flow control
  - Error control
  - Congestion control
- Flow control:
  - Matches sending and receiving data rates
  - Prevents receiver overload
- Error control:
  - Detects corrupted segments
  - Retransmits lost or damaged segments
- Congestion control:
  - Reduces packet loss due to network congestion

User Datagram Protocol (UDP)

- UDP stands for:
  - User Datagram Protocol
- UDP is:
  - Connectionless protocol
- UDP sends:
  - User datagrams independently
- No logical connection is established
- Each datagram:
  - Independent of previous datagrams
  - Independent of next datagrams
- UDP is:
  - Simple protocol
  - Low overhead protocol

Characteristics of UDP and SCTP

- UDP does not provide:
  - Flow control
  - Error control
  - Congestion control
- UDP suitable for:
  - Short messages
  - Fast communication
  - Applications avoiding retransmission delays
- SCTP stands for:
  - Stream Control Transmission Protocol
- SCTP designed for:
  - Emerging multimedia applications

Network Layer in TCP/IP

- Network layer provides:
  - Host-to-host communication
- Responsible for:
  - Creating connection between source and destination computers
- Communication path may contain:
  - Multiple routers
- Routers are responsible for:
  - Choosing best route for packets

Functions of the Network Layer

- Network layer responsibilities:
  - Host-to-host delivery
  - Packet routing
- Packets may travel through:
  - Multiple possible routes
- Network layer separates:
  - Routing tasks
  - Transport-layer tasks
- Routers do not require:
  - Application layer
  - Transport layer
- Separation reduces:
  - Router complexity
  - Number of required protocols

Internet Protocol (IP)

- Main network-layer protocol:
  - Internet Protocol (IP)
- IP defines:
  - Format of network-layer packet
- Network-layer packet is called:
  - Datagram
- IP also defines:
  - Address structure
  - Address format
- IP responsible for:
  - Routing packets from source to destination

Routing Process in IP

- Routing achieved through:
  - Routers forwarding datagrams
- Each router forwards packet to:
  - Next router in path
- IP is:
  - Connectionless protocol
- IP does not provide:
  - Flow control
  - Error control
  - Congestion control
- Applications requiring these services depend on:
  - Transport layer protocols

Routing Protocols

- Network layer includes:
  - Unicast routing protocols
  - Multicast routing protocols
- Unicast communication:
  - One-to-one communication
- Multicast communication:
  - One-to-many communication
- Routing protocols do not:
  - Perform routing directly
- Routing protocols create:
  - Forwarding tables for routers
- IP uses forwarding tables for:
  - Packet routing

Auxiliary Protocols in Network Layer

- Auxiliary protocols support:
  - IP delivery
  - IP routing
- ICMP:
  - Internet Control Message Protocol
  - Reports routing problems
- IGMP:
  - Internet Group Management Protocol
  - Supports multicasting
- DHCP:
  - Dynamic Host Configuration Protocol
  - Assigns network-layer addresses
- ARP:
  - Address Resolution Protocol
  - Finds link-layer address from network-layer address

Data-link Layer in TCP/IP

- An internet consists of:
  - Multiple LANs
  - Multiple WANs
  - Routers connecting links
- Multiple possible paths may exist:
  - Between source and destination
- Routers choose:
  - Best link for forwarding packets
- Data-link layer responsible for:
  - Moving datagram across selected link

Types of Links in Data-link Layer

- Data-link layer supports different link types:
  - Wired LAN
  - Wireless LAN
  - Wired WAN
  - Wireless WAN
- Wired LANs may include:
  - Link-layer switches
- Different links may use:
  - Different data-link protocols
- Regardless of link type:
  - Data-link layer transfers packets through the link

Protocols in the Data-link Layer

- TCP/IP does not define:
  - Specific data-link protocol
- TCP/IP supports:
  - Standard protocols
  - Proprietary protocols
- Requirement for a data-link protocol:
  - Must carry datagram through the link
- Data-link layer:
  - Encapsulates datagram into a frame
- Data-link packet is called:
  - Frame

Services Provided by Data-link Protocols

- Different data-link protocols provide:
  - Different services
- Some protocols provide:
  - Error detection
  - Error correction
- Some protocols provide:
  - Only error correction
- Data-link layer ensures:
  - Reliable transmission across a link

Physical Layer in TCP/IP

- Physical layer responsible for:
  - Carrying individual bits across the link
- Physical layer is:
  - Lowest layer in TCP/IP protocol suite
- Communication at physical layer is still:
  - Logical communication
- A hidden layer exists below physical layer:
  - Transmission media

Transmission Media

- Devices are connected through:
  - Transmission media
- Transmission media may be:
  - Cable
  - Air
- Transmission media does not carry:
  - Bits directly
- Transmission media carries:
  - Electrical signals
  - Optical signals

Bit Transmission in the Physical Layer

- Bits are received from:
  - Data-link layer
- Physical layer:
  - Transforms bits into signals
- Signals are transmitted through:
  - Transmission media
- Logical unit exchanged between physical layers:
  - Bit
- Multiple physical-layer protocols exist:
  - Convert bits into signals

Encapsulation and Decapsulation

- Important concept in protocol layering:
  - Encapsulation
  - Decapsulation
- Encapsulation:
  - Adding headers at each layer
- Decapsulation:
  - Removing headers at each layer
- Processes occur in:
  - Source host
  - Router
  - Destination host
- Link-layer switches:
  - Do not perform encapsulation or decapsulation

Encapsulation and Decapsulation

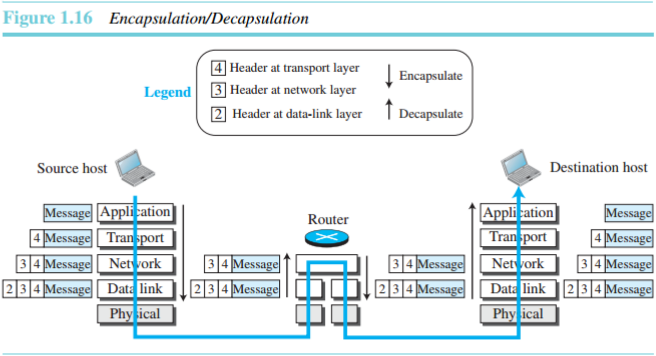


Figure: Encapsulation and Decapsulation Process

Encapsulation at the Source Host

- Source host performs:
  - Only encapsulation
- Data moves:
  - From higher layer to lower layer
- Each layer:
  - Adds its own header
- Result:
  - Data prepared for transmission

Application Layer Encapsulation

- Application-layer data called:
  - Message
- Message may contain:
  - Data only
  - Data with header or trailer
- Entire unit still called:
  - Message
- Message passed to:
  - Transport layer

Transport Layer Encapsulation

- Transport layer treats message as:
    - Payload
- Transport layer adds:
    - Transport-layer header
- Header contains:
    - Source application identifier
    - Destination application identifier
    - Flow control information
    - Error control information
    - Congestion control information
- Resulting packet:
    - Segment in TCP
    - User datagram in UDP
- Packet passed to:
    - Network layer

Network Layer Encapsulation

- Network layer treats transport packet as:
  - Payload
- Network layer adds:
  - Network-layer header
- Header contains:
  - Source host address
  - Destination host address
  - Error checking information
  - Fragmentation information
- Resulting packet called:
  - Datagram
- Datagram passed to:
  - Data-link layer

Data-link Layer Encapsulation

- Data-link layer treats datagram as:
  - Payload
- Data-link layer adds:
  - Link-layer header
- Header contains:
  - Link-layer source address
  - Link-layer destination address
- Resulting packet called:
  - Frame
- Frame passed to:
  - Physical layer

Encapsulation and Decapsulation at Router

- Router performs:
  - Decapsulation
  - Encapsulation
- Router connected to:
  - Multiple links
- Data-link layer:
  - Removes frame header
  - Extracts datagram
- Datagram passed to:
  - Network layer

Routing Operation at Router

- Network layer inspects:
  - Source address
  - Destination address
- Router consults:
  - Forwarding table
- Router determines:
  - Next hop
- Datagram normally:
  - Remains unchanged
- Exception:
  - Fragmentation if datagram too large
- Datagram sent to:
  - Data-link layer of next link

Re-encapsulation at Router

- Data-link layer of next link:
  - Encapsulates datagram into new frame
- New frame passed to:
  - Physical layer
- Physical layer:
  - Transmits bits through next link

Decapsulation at the Destination Host

- Destination host performs:
    - Decapsulation only
- Each layer:
    - Removes corresponding header
    - Extracts payload
- Payload delivered to:
    - Next higher layer
- Process continues until:
    - Message reaches application layer
- Decapsulation also includes:
    - Error checking

Addressing in the TCP/IP protocol suite

Figure 1.17 *Addressing in the TCP/IP protocol suite*

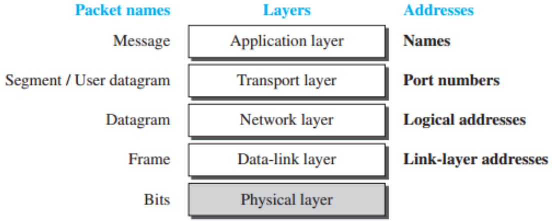


Figure: Addressing

Addressing in Protocol Layering

- Addressing is an important concept in:
  - Protocol layering
  - Internet communication
- Communication between two entities requires:
  - Source address
  - Destination address
- Logical communication exists between:
  - Corresponding protocol layers
- Four address pairs are normally used
- Physical layer does not require:
  - Addresses
- Reason:
  - Physical layer exchanges bits

Relationship Between Layers and Addresses

- Each layer has:
  - Specific address type
  - Specific packet name
- Address type depends on:
  - Function of the layer
- Different layers use:
  - Different addressing scopes

Application-layer Addressing

- Application layer uses:
  - Names
- Names identify:
  - Service providers
  - Users
- Examples:
  - someorg.com
  - somebody@coldmail.com
- Application-layer addresses are:
  - Human-readable

Transport-layer Addressing

- Transport-layer addresses are called:
    - Port numbers
- Port numbers identify:
    - Application programs
- Used at:
    - Source host
    - Destination host
- Port numbers are:
    - Local addresses
- Purpose:
    - Distinguish multiple running programs

Network-layer Addressing

- Network-layer addresses are:
  - Global addresses
- Scope:
  - Entire Internet
- Network-layer address uniquely identifies:
  - Device connection to Internet
- Used for:
  - Host-to-host delivery
  - Routing packets

Link-layer Addressing

- Link-layer addresses are called:
  - MAC addresses
- MAC stands for:
  - Media Access Control
- Link-layer addresses are:
  - Locally defined addresses
- Each address identifies:
  - Host in LAN or WAN
  - Router in LAN or WAN
- Used for:
  - Communication within a link

Addressing Summary

Layer	Address Type	Example
Application	Names	someorg.com
Transport	Port Numbers	80, 25
Network	IP Address	192.168.1.1
Data-link	MAC Address	00:1A:2B:3C:4D:5E
Physical	No Address	Bits/Signals

Addressing in the TCP/IP protocol suite

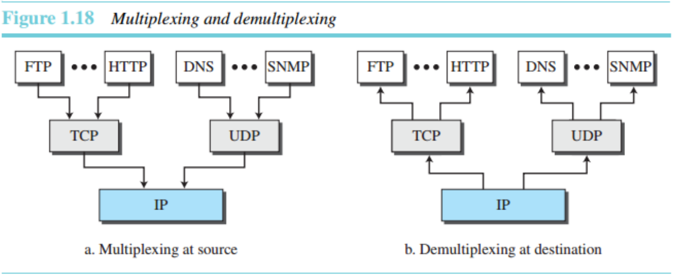


Figure: Multiplexing and demultiplexing

Multiplexing and Demultiplexing

- TCP/IP protocol suite uses:
  - Multiple protocols at some layers
- Source host performs:
  - Multiplexing
- Destination host performs:
  - Demultiplexing
- Multiplexing:
  - Encapsulation from multiple higher-layer protocols
- Demultiplexing:
  - Decapsulation and delivery to correct higher-layer protocol

Concept of Multiplexing

- A protocol at one layer can:
  - Accept packets from several higher-layer protocols
- Encapsulation occurs:
  - One packet at a time
- Example:
  - TCP accepts messages from different application protocols
- Multiplexing combines:
  - Multiple communication streams

Concept of Demultiplexing

- Demultiplexing occurs at:
  - Destination host
- A protocol can:
  - Decapsulate received packet
  - Deliver packet to appropriate higher-layer protocol
- Delivery occurs:
  - One packet at a time
- Demultiplexing separates:
  - Multiple communication streams

Need for Identification Fields

- Multiplexing and demultiplexing require:
  - Protocol identification field
- Identification field located in:
  - Packet header
- Field identifies:
  - Encapsulated higher-layer protocol
- Enables correct:
  - Packet delivery
  - Packet interpretation

Multiplexing at the Transport Layer

- Transport-layer protocols include:
    - TCP
    - UDP
- TCP and UDP can accept:
    - Messages from multiple application protocols
- Examples of application protocols:
    - HTTP
    - FTP
    - SMTP
- Transport layer multiplexes:
    - Application-layer messages

Multiplexing at the Network Layer

- IP accepts packets from:
  - TCP
  - UDP
  - ICMP
  - IGMP
- IP can receive:
  - Segments from TCP
  - User datagrams from UDP
- Network layer multiplexes:
  - Multiple transport and auxiliary protocols

Multiplexing at the Data-link Layer

- Data-link layer frame may carry:
  - IP packet
  - ARP packet
  - Other protocol packets
- Data-link layer identifies:
  - Type of encapsulated payload
- Supports communication between:
  - Different network-layer protocols

Summary of Multiplexing and Demultiplexing

Layer	Multiplexing From	Examples
Transport	Application Layer	HTTP, FTP, SMTP
Network	Transport Layer	TCP, UDP, ICMP
Data-link	Network Layer	IP, ARP

- Multiplexing combines packets from multiple protocols
- Demultiplexing delivers packets to correct protocols
- Header fields identify packet types

The OSI Model

- TCP/IP is not the only protocol suite
- Another important model:
    - OSI Model
- OSI stands for:
    - Open Systems Interconnection
- Developed by:
    - International Organization for Standardization (ISO)
- ISO established in:
    - 1947
- OSI model introduced:
    - Late 1970s

The OSI Model

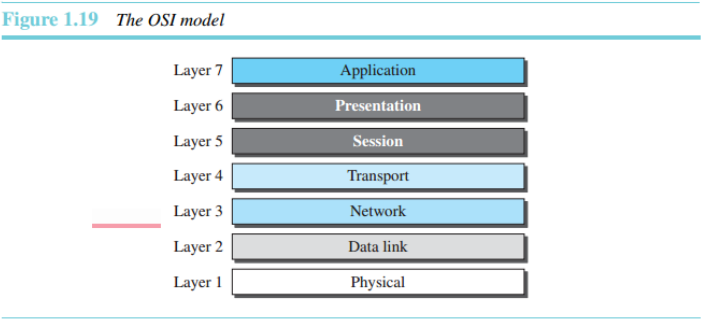


Figure: The OSI Model

International Organization for Standardization (ISO)

- ISO is:
  - Multinational standards organization
- Purpose of ISO:
  - Worldwide agreement on international standards
- Representation:
  - Nearly three-fourths of world countries
- ISO developed:
  - OSI communication model

Concept of Open Systems

- Open system:
  - Set of protocols enabling communication between different systems
- Communication possible regardless of:
  - Hardware architecture
  - Software architecture
- Goal of OSI model:
  - Facilitate communication between heterogeneous systems
- No changes required in:
  - Underlying hardware
  - Underlying software logic

Purpose of the OSI Model

- OSI model is:
  - Not a protocol
- OSI model provides:
  - Framework for network architecture
- Used for:
  - Understanding networks
  - Designing networks
- OSI model characteristics:
  - Flexible
  - Robust
  - Interoperable
- Intended as basis for:
  - OSI protocol stack

Layered Architecture of OSI

- OSI model uses:
  - Layered framework
- Supports communication between:
  - Different computer systems
- OSI model contains:
  - Seven layers
- Layers are:
  - Separate
  - Related
- Each layer defines:
  - Part of information transfer process

Seven Layers of the OSI Model

Layer Number	OSI Layer
7	Application Layer
6	Presentation Layer
5	Session Layer
4	Transport Layer
3	Network Layer
2	Data-link Layer
1	Physical Layer

OSI Model versus TCP/IP

- TCP/IP model has:
  - Fewer layers than OSI model
- Missing OSI layers in TCP/IP:
  - Session layer
  - Presentation layer
- TCP/IP application layer combines:
  - Application layer
  - Presentation layer
  - Session layer
- OSI application-related layers merged into:
  - Single application layer in TCP/IP

TCP/IP and OSI model

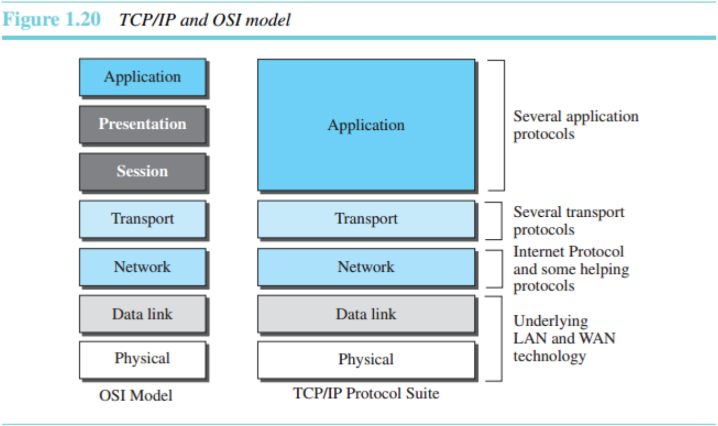


Figure: TCP/IP and OSI model

Reasons for Missing Layers in TCP/IP

- Reason 1:
  - TCP/IP has multiple transport protocols
- Some session-layer functions available in:
  - Transport-layer protocols
- Reason 2:
  - Application layer supports many applications
- Session and presentation functions can be:
  - Implemented within application software
- TCP/IP provides:
  - Greater application flexibility

OSI vs TCP/IP Layer Mapping

OSI Model	TCP/IP Model
Application	3*Application
Presentation	
Session	
Transport	Transport
Network	Network
Data-link	Data-link
Physical	Physical